

CY1010_Exam1_StudyGuide

CY1010 Environmental Chemistry — Exam 1 Study Guide

Covers: Part 1 (Dr. Anup Bhunia) + Part 2 (Dr. Abhijit Sau) — Lectures 1–9 **Exam 1:** 5 Sept 2026, 50 marks, MCQ, 90 min

PART 1 — Dr. Anup Bhunia

1. Earth & the Environment

- Environment = atmosphere–water–earth surroundings that make life possible.
- Earth is unique vs Mars/Venus: has liquid water, ~21% O₂, avg temp 15°C, supports life.
- Photosynthesis created Earth's oxygen atmosphere.
- **Lithosphere:** continental + oceanic crust + solid upper mantle, resting on the partially molten asthenosphere.
- Without atmosphere, Earth's avg temp would be **–18°C** instead of +15°C (greenhouse effect).
- Atmosphere composition: **N₂ 78%, O₂ 21%, Ar 0.93%, CO₂ ~0.04%**.

2. Water

- Only **2.5%** of Earth's water is freshwater.
- Salinity = total dissolved salt (TDS): ocean water ~33–37 g/L; drinking water <300 mg/L.
- Measured in ppm/ppb.
- **Classification of water by salinity** (know the ranges!):

Water type	Conductivity (dS/m)	Salt conc. (mg/L)	Example
Non-saline	<0.7	<500	Drinking, irrigation, surface water
Slightly saline	0.7–2	500–1,500	Surface/irrigation/ground water
Moderately saline	2–10	1,500–7,000	Runoff and groundwater
Highly saline	10–25	7,000–15,000	Saline groundwater/runoff
Exceptionally saline	25–45	15,000–35,000	Very saline groundwater, seawater
Brine	>45	>45,000	Saline inland lakes/seas

3. Nitrogen Cycle

- N_2 (78% of atmosphere) is unusable directly by most organisms.
- Cycles via: **nitrogen fixation** → **ammonification** → **nitrification** → **assimilation** → **decomposition** → **denitrification**.

4. Fossil Fuels

- Formed by: (1) photosynthesis > respiration, (2) detritus accumulation, (3) burial, (4) heat & pressure (metamorphosis).
- Combustion: $C_xH_y + (x+y/4)O_2 \rightarrow xCO_2 + (y/2)H_2O + \text{heat}$.
- Reserves: Oil & gas ~50 yr; Coal ~380 yr.
- **Fractional distillation** of crude oil separates by boiling point (C_1 – C_5 gases → ... → > C_{70} residue).
- **LPG**: mainly propane + butane; odourless; mercaptan added for leak detection; 220–2200 kPa.
- **Natural gas**: mainly methane; CNG compressed at 20–25 MPa; mercaptan added.
- **Coal types**: Anthracite (90% C, hard) > Bituminous (80% C, soft) > Lignite (70% C, brown). Coal reserves ~200 yrs (or 265 yrs if usage grows 2%/yr). India has 5% of world coal but high ash content.

5. Hydrogen Energy

- Fuel cell: Anode splits $H_2 \rightarrow H^+ + e^-$ (Pt catalyst); Cathode: $O_2 + H^+ + e^- \rightarrow H_2O$.
- **Steam Methane Reforming (SMR)**: $CH_4 + H_2O \rightarrow CO + 3H_2$ (~50% of world's H_2).
- **Coal gasification**: $3C + O_2 + H_2O \rightarrow H_2 + 3CO$ (~20% of world's H_2).
- **Water electrolysis**: $H_2O \rightarrow H_2 + \frac{1}{2}O_2$ (green electricity) (~4% of world's H_2).
- **Types of H_2 (memorize!)**:

	Type	Source
	Grey	SMR (natural gas)
	Black/Brown	Coal gasification
	Blue	SMR + carbon capture & storage
	Green	Electrolysis with renewable electricity

- **Calorific values of fuels (MJ/kg)** — Hydrogen has by far the highest:

Fuel	Calorific value
Hydrogen (H_2)	120–142 MJ/kg (highest)
Natural gas	42–55
Methane (CH_4)	50–55

Fuel	Calorific value
LPG	46–51
Crude oil	42–47
Petrol/gasoline	44–46
Diesel	42–46
Hard black coal	>23.9
Methanol (CH ₃ OH)	22.7
Dimethyl ether (DME)	29
Sub-bituminous coal	17.4–23.9
Lignite/brown coal	<17.4
Firewood (dry)	16

6. Solar, Wind, Nuclear

- Sun's power: 384 Yotta Watt (3.84×10^{26} W); intensity at Earth's orbit $\approx$ **1377 W/m²**.
- Earth receives $\sim$ 5.5 million EJ/year from sun; humankind uses $\sim$ 500 EJ/year $\rightarrow$ entire annual human energy use = energy received from sun in **$\sim$ 1 hour**.
- Solar uses: Solar thermal (water heater, cooker, furnace) vs Photovoltaic.
- Wind: caused by (1) uneven heating $\rightarrow$ convection, (2) Earth's rotation. $\sim 1.6 \times 10^7$ MW available globally; 2025: wind supplied $\sim$ 2700 TWh ($\sim$ 8% of world electricity).
- **Nuclear fission:** $\text{U-235} + \text{n} \rightarrow \text{Ba-144} + \text{Kr-89} + 3\text{n} + \text{Energy}$.
- **Nuclear fusion:** ${}^2\text{H} + {}^3\text{H} \rightarrow {}^4\text{He} + \text{n} + \text{Energy}$.
- 4 Indian nuclear plants: Tarapur, Rana Pratap Sagar, Kalpakkam, Narora.
- Seawater contains $\sim$ 4 billion metric tons of uranium (continuously replenished) $\rightarrow$ arguably renewable.
- **Pumped hydro:** dominant (98%) grid-scale energy storage; stores $\sim$ 3 Wh per m³ per 1 m lift.
- **EV energy storage** — benefits: no pollution/noise, high torque & power, up to 18,000 rpm (vs $\sim$ 6,000 rpm mechanical engine), regenerative braking, lower maintenance. Issues: limited range (range anxiety), long charging time, high initial cost, battery lifetime.
- **Fuel cell types:** Low-temperature ($<200^\circ\text{C}$, alkaline/acidic electrolyte, uses H₂ gas) vs High-temperature ($>450^\circ\text{C}$, uses hydrocarbon fuel like methane/gasoline directly).
- Hydrogen fuel cells also used in aerospace: Helios unmanned aircraft (solar-regenerated H₂ fuel cell); Airbus "ZEROe" concepts (2020) targeting first zero-emission commercial aircraft by 2035.

7. Biomass & Biofuels

- Biomass: renewable organic material from living organisms.
 - Edible: cereals, starch, sugar, oil crops.

- Non-edible (lignocellulosic): agricultural/forest residues.
- Global biomass production >10 billion tons/yr (~10% of global energy supply).
- **Starch vs Cellulose**: both glucose polymers; starch is linear/branched with intra+inter H-bonds (edible); cellulose is linear with only intra H-bonds (non-edible).
- **Biogas**: 55–65% methane, 30–40% CO₂, rest H₂/H₂S/N₂.
- **Geothermal**: steam wells up to 1000 m, 200–300°C; needs moisture/particle separation before turbine.
- **OTEC** (Ocean Thermal Energy Conversion): uses temperature gradient (surface vs deep water, 20–25°C difference); low-boiling fluid (ammonia/propane) drives turbine.
- **Tidal energy**: flood tide (above mean sea level) vs ebb tide (below); Sihwa Lake (S. Korea) = world's largest tidal plant (254 MW).

Biofuel generations:

Gen	Source	Product
1st	Starch/sugarcane	Bioethanol, biodiesel
2nd	Lignocellulosic waste (sawdust, corncob, bagasse)	Bioethanol (carbon neutral)
3rd	Algae	Biodiesel (carbon neutral)
4th	Genetically engineered microorganisms	Biodiesel/bioethanol (carbon negative , absorb 30% more CO ₂)

- **Transesterification**: Triglyceride + 3 MeOH → (base catalyst) → Biodiesel (FAME) + Glycerol.
- Blending: **Biodiesel/Ethanol** % — B100, B20, B7, B5, B2. India target: B5 by 2030 (biodiesel), E20 by 2030 (bioethanol).

8. Catalysis

- Catalyst: speeds up reaction by lowering activation energy (E_a), not consumed.
- Examples: NaOH (biodiesel), Ni (SMR), Fe (Haber-Bosch), Ti/Al catalysts (polymer synthesis - Ziegler-Natta, Nobel 1963).
- **Haber-Bosch**: N₂ + 3H₂ → (Fe) → 2NH₃. Most energy-intensive industrial process (1.5–2% of global energy). 3 Nobel Prizes: Haber (1918), Bosch/Bergius (1931), Ertl (2007).
- **Heterogeneous catalysis** steps: adsorption → bond breaking → surface reaction → desorption.
 - Example: C₂H₄ + H₂ → (Ni/Pt/Pd) → C₂H₆.
- **Nitrogenase** enzyme (in root-nodule bacteria) fixes N₂ → NH₃ biologically.
- **Ostwald Process** (nitric acid): 4NH₃ + 5O₂ → (Pt) → 4NO + 6H₂O; 2NO + O₂ → 2NO₂; 2NO₂ + H₂O → HNO₂ + HNO₃.

- **Catalytic converters:** $\text{CO} + \text{hydrocarbons} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$; $2\text{NO} + 2\text{NO}_2 \rightarrow 2\text{N}_2 + 3\text{O}_2$.
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PART 2 — Dr. Abhijit Sau

9. Microbial Degradation of Organics

- **Biodegradation:** breakdown of organic matter by microorganisms.
- **Bioremediation:** using organisms (natural/introduced) to break down pollutants $\rightarrow \text{CO}_2$, H_2O , biomass.
- Requires: right pollutant type, solubility, bioavailability, favorable microbial conditions.
- Oil spill causes: tanker accidents, offshore drilling, pipeline leaks, equipment failure, natural disasters.
- **Oil spill remediation methods:**
 1. In-situ burning — fast, but produces air pollutants.
 2. Mechanical — booms, skimmers, vacuum trucks (no chemical change).
 3. Chemical — dispersants/surfactants (risk of secondary pollution).
 4. Biological — bacteria/fungi degrade oil (eco-friendly, cheap).
- Biodegradation of oil occurs at oil–water interface, needs O_2 ; warmer climate favors it.

Enzymes in microbial degradation: Oxygenases (mono/di-oxygenases), Dehydrogenases, Hydrolases, Peroxidases, Laccases (fungi).

Metabolic pathways intermediates enter: β -oxidation (fatty acids/hydrocarbons), TCA (citric acid) cycle, Glycolysis, Electron transport chain.

- **β -oxidation:** carboxyl group at C-1 activated via coenzyme A $\rightarrow$ stepwise oxidation at C-3 (β -position).
- Detergent/petroleum breakdown: terminal oxidation of hydrocarbon chain $\rightarrow$ carboxylic acid $\rightarrow$ β -oxidation; catalyzed by **cytochrome P-450** (a monooxygenase).
- Linear hydrocarbons/detergents degrade similarly to soaps (fast); cyclic/aromatic compounds also oxidized via cytochrome P-450 but more slowly (esp. high MW compounds — poor cell-wall penetration).

10. Oxidation of Aromatic Compounds (know these transformations!)

- **Benzene $\rightarrow$ Maleic anhydride:** benzene + air, catalyst, 2–5 bar, 400–450°C. Byproduct: fumaric acid.
- **Naphthalene** (2 fused benzene rings, from coal tar, mp 80°C, bp 218°C) oxidation products — **exact reagent $\rightarrow$ product pairing (commonly missed on MCQs, memorize precisely):**

Reagent	Product
KMnO ₄ / Acidic	Phthalic Acid
KMnO ₄ / Basic	Phthalonic Acid
H ₂ Cr ₂ O ₇	1,4-Naphthoquinone
O ₃ , Zn/H ₂ O	Phthalaldehyde
O ₂ , V ₂ O ₅	Phthalic Anhydride

- **Methanol oxidation** → first metabolite = **Formaldehyde** (causes neuronal damage/toxicity).
- **Benzoic acid + Glycine** → **Hippuric acid** (via benzoyl-CoA activation in liver/intestine/kidney) — a detox pathway.

11. Polymers & Environment

- Polyethylene + 5–20% starch granules → used for "biodegradable" bags.
- Transition-metal additives can enhance photodegradation of polyethylene via hydroperoxides.
- PHVB copolymer can be produced by microorganisms fed β-hydroxyvalerate, or by genetically engineered plants (oilseed rape, 1999) — not yet economical.
- Fun scale fact: total bacterial + archaeal cells on Earth ≈ **1×10³⁰**; largest habitats are deep oceanic subsurface, soil, and deep continental subsurface (each ~10²⁹).
- Organic chemicals in environment: pharmaceuticals, synthetic fibers, building materials, agrochemicals, solvents, cleaning agents.
- **12 Key POPs (Persistent Organic Pollutants)** include PCBs (polychlorinated biphenyls) — persistent, bioaccumulative, toxic.

12. Greenhouse Gases & Global Warming (calculation-heavy — practice these!)

- Volume of Earth ≈ 1.083×10²¹ m³; Troposphere ≈ 6.13×10¹⁸ m³ (~80% of atmosphere's mass).
- CO₂ in troposphere ≈ 2.453×10¹⁵ m³ (at 0.04%).
- World energy use ≈ 500 EJ/yr; 80–85% from fossil fuels.
- C + O₂ → CO₂, ΔH ≈ −400 kJ/mol.
- Moles of CO₂ produced/yr ≈ 5×10²⁰ J ÷ 4×10⁵ J/mol ≈ **1.25×10¹⁵ mol**.
- Volume of that CO₂ at STP (22.4 L/mol) ≈ **2.83×10¹³ m³**.
- **Scattering vs Absorption**: visible light scattered by O₂/N₂ (Rayleigh scattering → blue sky); IR absorbed by GHGs.
- **Atmospheric window**: portion of EM spectrum transmitted through atmosphere without absorption.

Key greenhouse gases:

Gas	Notes	GWP	Lifetime
CO ₂	Baseline	1	Variable
CH ₄ (methane)	From anaerobic decomposition, rice paddies, coal mines; pre-industrial 0.7 ppm → now ~1.8 ppm	25	12.2 yr
N ₂ O	From denitrification & fertilizer use & combustion; pre-industrial 270 ppb → 2021: 340 ppb (+20%)	298	120 yr
HFCs	No Cl, don't harm ozone but absorb IR (warming); broken down by ·OH in troposphere	12–14,800	1.5–264 yr
PFCs	Byproducts of Al production & semiconductor mfg (e.g. CF ₄ , C ₂ F ₆)	7,390–12,200	3200–50000 yr
SF ₆	Most potent GHG; used as electrical insulator; very stable (paired electrons)	22,800	3200 yr
CFCs	Break down stratospheric ozone; broken only by UV (release Cl)	—	—
O ₃	Stratospheric (good, blocks UV) vs Tropospheric (bad, forms from NO _x +VOC+sunlight, traps heat)	—	—

- Sulphate particles: from SO₂ oxidation (coal/oil combustion) → absorb IR + cause **acid rain** (H₂SO₄).
- Carbon particles (diesel engines): absorb heat & reflect light; short residence time (~2 weeks) unlike gases.
- **Paris Agreement (2015)**: limit warming well below 2°C, preferably 1.5°C above pre-industrial levels.
- Shifting coal → natural gas: methane burns cleaner, emits ~60% less CO₂ per unit electricity than coal; less particulate matter; coal has 0.5–5% S (SO₂/acid rain) and 0.5–3% N (NO_x).

13. Plastic Pollution & Carbon Capture

- Plastic pros: durable, cheap, light. Cons: microplastic leaching, bioaccumulation in aquatic life.
- Plastic-to-fuel: Pyrolysis, thermal decomposition, catalytic converters (zeolites, cobalt).
- **DACS** (Direct Air Capture & Storage): CO₂ from ambient air. **CCS** (Carbon Capture & Storage): CO₂ from point sources (power plants).
- DAC capturing agents: (1) Liquid solvents (NaOH/KOH → carbonates → CaCO₃ precipitate; regenerate at ~900°C) or (2) Solid sorbents (MOFs, zeolites, porous silica, amine-functionalized — regenerate at 50–120°C).
- **Physisorption vs Chemisorption**:

		Physisorption
	Bonding	Weak, van der Waals
	Surface-specific?	No
	ΔH_{ads}	5–50 kJ/mol
	Temperature effect	Higher T reduces coverage

- **Pore size (IUPAC):** Micropores <2 nm (zeolite, activated C, MOF); Mesopores 2–50 nm; Macropores >50 nm.
- **MOFs** (Metal-Organic Frameworks) — 2025 Nobel Prize in Chemistry (Kitagawa, Robson, Yaghi); up to 94% empty space; tunable pores.
- **CO₂ utilization for chemical synthesis** (~130 Mt CO₂/yr):
 - Urea: $\text{CO}_2 + \text{NH}_3 \rightarrow \text{urea}$ (150–250°C, 2 MPa) — biggest CO₂ consumer.
 - Salicylic acid: Kolbe-Schmitt process (phenol + NaOH → sodium phenoxide + CO₂ → sodium salicylate → acidify → salicylic acid).
 - Polycarbonates: ring-opening copolymerization of epoxides + CO₂ (ZnEt₂ catalyst) → CDs, lenses, auto parts.
- Global carbon budget: temp rise by 2100 could be >4°C without action; goal 1.5–2°C. DAC = only negative-emission technology.
- **Photocatalysis** steps: (I) light absorption → e⁻/hole pairs, (II) charge separation, (III) transfer to surface, (IV) redox reactions.
- UN Sustainable Development Goals (SDGs) — 17 goals for people & planet.
- **3R:** Reduce (source reduction) → Reuse → Recycle.
- Environmentalists to know: Wangari Maathai (Kenya, Nobel Peace 2004, Green Belt Movement), M.C. Mehta (India), Rajendra Singh ("Waterman of India"), Salim Ali ("Birdman of India"), Sunderlal Bahuguna (Chipko movement).

14. Green Chemistry & Industrial Ecology

- **Definition:** set of principles that reduce/eliminate hazardous substances in design, manufacture, and application of chemical products.
- **Pollution Prevention Act (PPA) hierarchy:** Prevent → Recycle → Treat → Dispose (last resort).
- Green chemistry = pollution prevention at the molecular scale.

12 Principles of Green Chemistry (Anastas & Warner, 1998) — MEMORIZE:

1. Prevention (prevent waste, don't clean up after)
2. Atom Economy (maximize incorporation of materials into final product)
3. Less Hazardous Chemical Synthesis
4. Designing Safer Chemicals
5. Safer Solvents & Auxiliaries
6. Design for Energy Efficiency (ambient T/P preferred)

7. Use of Renewable Feedstocks
8. Reduce Derivatives (avoid unnecessary protection/deprotection)
9. Catalysis (catalytic > stoichiometric reagents)
10. Design for Degradation (products break down harmlessly)
11. Real-time Analysis for Pollution Prevention
12. Inherently Safer Chemistry for Accident Prevention

Established Green Chemistry Metrics (formulas — likely calculation MCQs):

1. **Atom Economy** = (Mass of desired product / Mass of all reactants) × 100 **(including stoichiometric coefficients)*
2. **E-factor (Environmental factor)** = Total waste (g) / Product (g) — lower is better; ideal is 0.
3. **% Atom Utilization** = (MW of desired product / MW of all products) × 100
4. **Reaction Mass Efficiency (RME)** = (mass of product C / (mass of A + mass of B)) × 100
5. **% Yield** = (Actual Yield / Theoretical Yield) × 100

Worked atom-economy examples:

- **Case 1 (Wittig reaction):** PhC(Me)=O (MW 120) + $\text{H}_2\text{C=PPh}_3$ (MW 276) → PhC(Me)=CH_2 (MW 118, desired) + O=PPh_3 (MW 278, waste). Atom economy = $118 / (120+276) \times 100 = 29.79\%$ (substitution-type reaction → poor atom economy, large waste byproduct).
- **Case 2 (Hydroformylation):** alkene + CO + H₂ → (catalyst) → aldehyde (linear or branched). Atom economy = **100%** — nothing is wasted because it's an **addition reaction** (all atoms of reactants end up in the product). → **Key insight:** addition reactions tend to have high/perfect atom economy; substitution/elimination reactions tend to waste atoms as byproducts.
- **Ibuprofen synthesis — classic green chemistry case study:**
 - Original **Boots process** (1960s): 6 steps, uses stoichiometric reagents (AlCl₃, NaOC₂H₅, NH₂OH, etc.), poor atom economy, large waste byproducts.
 - Greener **BHC process** (1991, only 3 steps): uses HF (recoverable catalyst), Raney Ni/H₂, and Pd/CO catalytically — most atoms of reactants end up in the final ibuprofen product, much less waste. This is the textbook example of applying green chemistry principles (atom economy, catalysis, waste reduction) to redesign an industrial synthesis.
- Epoxidation of styrene: Atom Economy = 23% (77% waste as m-chlorobenzoic acid byproduct).
- Phenol production: conventional (from petroleum, unsustainable) vs alternative (from biomass waste via depolymerization — uses an abundant waste stream as feedstock).
- Ethylene oxide: greener process (uses O₂ instead of Cl₂) generates **16× less waste** than the original, and eliminates wastewater formation.

- Less hazardous synthesis tools: Cascade/tandem reactions, C–H activation, Metathesis, Enzymatic reactions (e.g., lipase-catalyzed).

15. GHS Hazard Communication (know categories!)

- **Physical hazards:** explosives, flammable gases/liquids/solids/aerosols, oxidizers, gases under pressure, pyrophorics, self-reactive/self-heating substances, organic peroxides, corrosive to metals.
 - **Health hazards:** acute toxicity, skin/eye corrosion-irritation, respiratory/skin sensitization, mutagenicity, carcinogenicity, reproductive toxicity, target organ toxicity, aspiration toxicity.
 - **Environmental hazards:** aquatic toxicity (acute/chronic), bioaccumulation potential, rapid degradability.
 - **Label elements:** Product identifier, Signal word (Danger=severe / Warning=less severe), Hazard statement, Pictogram(s), Precautionary statement, Manufacturer info.
 - **Pictograms** (black symbol, white background, red diamond frame): Health hazard (carcinogen etc.), Environment (aquatic toxicity), Exclamation mark (irritant), Skull & crossbones (acute toxicity), Gas cylinder (pressure), Corrosion, Exploding bomb (explosives), Flame over circle (oxidizers), Flame (flammables).
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Quick Formula/Number Cheat-Sheet

- Earth avg temp: 15°C (would be –18°C without atmosphere)
- Atmosphere: N₂ 78%, O₂ 21%, Ar 0.93%, CO₂ 0.04%
- Freshwater: 2.5% of total water
- Solar constant at Earth's orbit: 1377 W/m²
- Sun's power output: 3.84×10²⁶ W
- Global energy use: ~500 EJ/yr
- CO₂ moles produced/yr: ~1.25×10¹⁵ mol
- SF₆ GWP: 22,800× CO₂ (highest)
- CH₄ GWP: 25; N₂O GWP: 298
- SMR gives 50% of world H₂; coal gasification 20%; electrolysis 4%
- Micropores <2nm, Mesopores 2–50nm, Macropores >50nm
- Physisorption ΔH: 5–50 kJ/mol; Chemisorption ΔH: 50–500 kJ/mol
- Paris Agreement: <2°C, ideally 1.5°C
- Haber-Bosch: 1.5–2% of global energy consumption
- H₂ calorific value: 120–142 MJ/kg (highest of all common fuels)
- Naphthalene + O₂/V₂O₅ → Phthalic Anhydride (NOT naphthoquinone — that's H₂Cr₂O₇)
- Atom economy formula: (mass desired product / mass all reactants) × 100; addition reactions ≈ 100%, substitution reactions often <30%

- E-factor = waste(g)/product(g); Ibuprofen: Boots process (6 steps, wasteful) vs BHC process (3 steps, atom-economical)
 - Total bacteria+archaea on Earth: $\sim 1 \times 10^{30}$ cells
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How to Use This Guide

1. Read through once fully.
2. Focus extra time on: **12 Principles of Green Chemistry, Types of H₂, GHG table (GWP/lifetime), naphthalene/benzene oxidation products**, and the **CO₂ calculation numbers**.
3. Ask me to quiz you — I can generate a practice MCQ quiz from this material any time.